Status	Finished
Started	Monday, 5 October 2026, 9:02 AM
Completed	Monday, 5 October 2026, 9:05 AM
Duration	3 mins 5 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O👎
b. O(1)
c. O(log n) ✓
d. O(n log n)

Question **2**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By comparing each element in the list sequentially ✓
c. By dividing the list into halves
d. By sorting the list first

Question **3**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element
b. Binary search works on unsorted lists
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search does not require dividing the list

Question **4**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O👎
c. O(1) ✓
d. O(log n)

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search stops immediately ✓
b. The search starts over from the beginning
c. The search continues until the end of the list
d. The search backtracks to find duplicate elements

Question **6**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and unsorted
b. When the list is small and sorted
c. When the list is small and unsorted ✓
d. When the list is large and sorted

Question **7**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 100
c. 1000
d. 500

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The search continues from the beginning
- c. The search stops
- d. The list is sorted

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(1)
- b. O(log n)
- c. O(n log n)
- d. O👎 ✓

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Quick search
- b. Merge search
- c. Bubble search
- d. Linear search ✓

Question **11**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n-1
- c. n/2
- d. log n

Question **12**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Binary search
c. Jump search
d. Linear search ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining
b. Finding
c. Searching ✓
d. Discovering

Question **14**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all
b. When the item is the last element in the array
c. When the item is not the array at all
d. When the item is somewhere in the middle of the array ✓

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. There must be a mechanism to delete and/or insert elements in the list ✓

c. There should be direct access to the middle element in any sublist

d. The list must be sorted